Evaluating AI Models for Business Name Detection in Mobile Databases

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1. INTRODUCTION

Digital forensic investigators analyze mobile application databases to find important information such as business names, organizations, and communication records.

Manually searching through database tables and columns is:

- Time-consuming
- Labor-intensive
- Difficult to scale

Identifying business names in text is a type of **Named Entity Recognition (NER)**, a common task in natural language processing (Wang et al., 2025).

Recent research shows that large language models (LLMs) can perform NER tasks, but their performance depends on data quality and structure (Zhao, 2025).

AI models may help automate this process, but their accuracy in structured forensic database data is still being studied.

2. OBJECTIVE

The goal of this project is to evaluate how accurately AI models can identify business names in mobile database data using a manually created ground truth.

3. DATASET

Seven mobile forensic databases were analyzed:

- A1_msgstore.db
- I1_ChatStorage.sqlite
- A1_wa.db
- I3_sms.db
- I4_CloudTabs.db
- I4_History.db
- A5_SBrowser.db

Each database was converted into a CSV file for analysis.

4. GROUND TRUTH

Business names were manually identified by:

- Opening each database file
- Searching tables, columns, and rows
- Recording all visible business names

A ground-truth TXT file was created for each database.

Business Names Identified per Database

Database	Number of Business Names
A1_msgstore.db	9
I1_ChatStorage.sqlite	7
A1_wa.db	9
I3_sms.db	8
I4_CloudTabs.db	5
I4_History.db	8
A5_SBrowser.db	1
Total	**47**

A total of 47 business names were manually identified across all databases and used as ground truth for evaluation.

5. AI MODELS TESTED

•ChatGPT GPT-5.3
•Grok 4
•Microsoft 365 Copilot GPT-5

6. METHODOLOGY

The following steps were performed for each database:

1. The database file was opened and manually analyzed
2. A ground-truth list of business names was created
3. The database was converted into a CSV file
4. The CSV file was reviewed to verify all ground-truth names were present
5. The CSV file and ground-truth list were given to the AI models

The AI models were instructed to:

- Search for each business name in the CSV
- Match names case-insensitively
- Only use the provided list (no guessing)

Each model returned:

- **FOUND** → names detected
- **NOT FOUND** → names missed

These results were compared with the ground truth.

Ground-truth business names were verified in the CSV files after conversion to ensure no data loss before evaluating AI models.

This evaluation approach follows common practices in Named Entity Recognition tasks, where model outputs are compared against manually labeled data (Wang et al., 2025).

7. EVALUATION METRICS

Precision
Precision = TP / (TP + FP)
When the AI says it found a name, how often is it correct?
Recall
Recall = TP / (TP + FN)
Out of all real business names, how many did the AI find?
F1 Score
F1 = 2 × (Precision × Recall) / (Precision + Recall)
A single score that balances precision and recall.
Note on Accuracy
Accuracy was not used because the dataset did not include true-negative examples.
TP = correctly found names, FP = incorrect matches, FN = missed names.

Workflow Overview

This project followed a step-by-step process to evaluate AI performance:

1. Open mobile database files (SQLite)
2. Manually identify business names (ground truth)
3. Convert databases into CSV files
4. Verify ground-truth names in the CSV files
5. Provide CSV and ground truth to AI models
6. AI models check which names appear in the data
7. Collect results (FOUND / NOT FOUND)
8. Compare results with ground truth
9. Calculate performance metrics

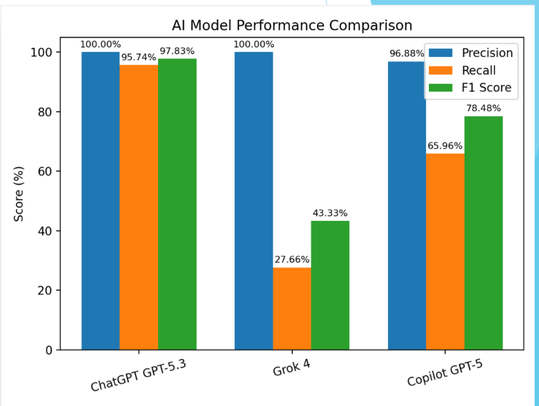
Short Workflow:
Database → Ground Truth → CSV → Verification → AI → Results → Comparison → Metrics

8. RESULTS

Model	TP	FP	FN	Precision	Recall	F1 Score
ChatGPT GPT-5.3	45	0	2	100.00%	95.74%	97.83%
Grok 4	13	0	34	100.00%	27.66%	43.33%
Copilot GPT-5	31	1	16	96.88%	65.96%	78.48%

TP = True Positive (correctly found business names), FP = False Positive (incorrect matches), FN = False Negative (missed names).

Model Performance Comparison



9. ERROR ANALYSIS

Some AI models missed business names due to:

- Conservative detection (only selecting high-confidence matches)
- Noisy and unstructured database data
- Large CSV files causing missed entries
- Formatting and encoding differences

Research shows that LLMs can struggle with structured and domain-specific data, especially when the data is noisy or inconsistent (Lu & Huo, 2025).

This explains why some models had high precision but low recall.

10. FUTURE WORK

To improve performance:

- Clean and normalize CSV data
- Split large files into smaller sections
- Improve prompt design
- Train models on forensic-specific data
- Combine AI with rule-based methods

11. CONCLUSION

ChatGPT GPT-5.3 achieved the best overall performance.
AI models can assist digital forensic investigations, but performance depends on:

- data quality
- model behavior
- data structure

12. REFERENCES

1. Wang, S., Liu, Y., & Chen, Z. (2025). *GPT-NER: Named Entity Recognition via Large Language Models.* Proceedings of the Association for Computational Linguistics (ACL).
2. Lu, Y., & Huo, Y. (2025). *Financial Named Entity Recognition: How Far Can Large Language Models Go?* arXiv preprint arXiv:2501.02237.
3. Zhao, Y. (2025). *Large Language Models for Named Entity Recognition: A Survey.* Artificial Intelligence Review.